49. Boundary Scan

49.1 Overview

The boundary scan function provides a serial I/O interface based on the JTAG (Joint Test Action Group), IEEE Std.1149.1, and IEEE Standard Test Access Port and Boundary Scan Architecture. Table 49.1 lists the boundary scan specifications, Figure 49.1 shows a block diagram, and Table 49.2 lists the I/O pins.

Table 49.1 Boundary scan specifications

Parameter	Specifications
Execution condition	Boundary scan must be executed when the RES pin is driven low.
Test modes	● BYPASS mode ● EXTEST mode ● SAMPLE/PRELOAD mode ● CLAMP mode ● HIGHZ mode ● IDCODE mode

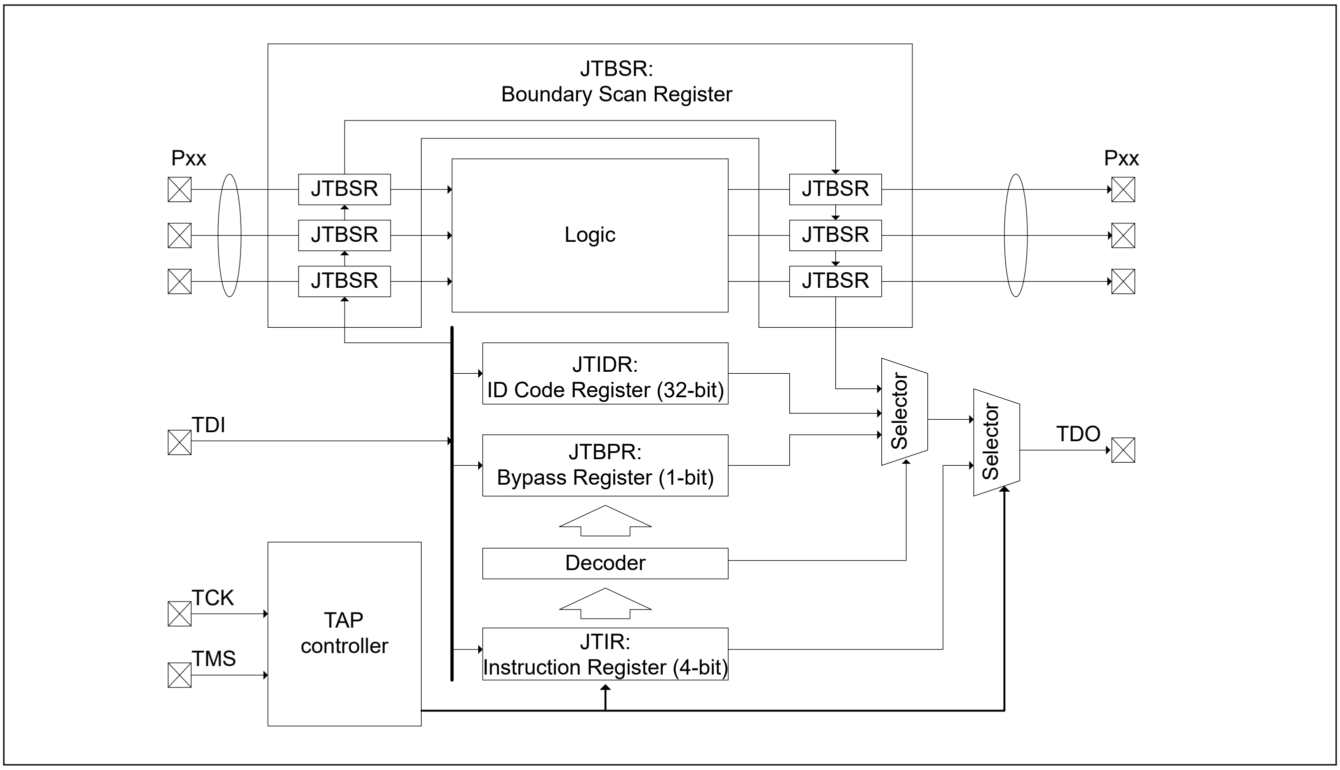

Figure 49.1 Boundary scan function block diagram

Table 49.2 Boundary scan I/O pins

Pin name	I/O	Description
TCK	Input	Test clock input pin Clock signal for boundary scan. The input clock duty cycle is 50% when the boundary scan function is used.
TMS	Input	Test mode select pin
TDI	Input	Test data input pin
TDO	Output	Test data output pin

Note: This device does not support the TRST pin for the JTAG interface.

49.2 Register Descriptions

Table 49.3 lists the boundary scan registers.

Table 49.3 Boundary scan registers

Register name	Symbol	Value after reset
Instruction Register	JTIR	0xE
ID Code Register	JTIDR	0x085D_A447
Bypass Register	JTBPR	Undefined
Boundary Scan Register	JTBSR	Undefined

Usage notes for the boundary scan registers:

- Instructions can be input to the Instruction Register (JTIR) through the TDI pin by serial transfer.
- The Bypass Register (JTBPR), which is a 1-bit register, is connected between the TDI and TDO pins in BYPASS mode.
- The Boundary Scan Register (JTBSR), which is configured according to the BSDL description, is connected between the TDI and TDO pins when test data is being shifted in.

Table 49.4 shows the availability of serial transfer for the registers.

Table 49.4 Serial transfer for registers

Register name	Serial input	Serial output
Instruction Register (JTIR)	Available	Available
ID Code Register (JTIDR)	Available	Available
Bypass Register (JTBPR)	Available	Available
Boundary Scan Register (JTBSR)	Available	Available

49.2.1 JTIR : Instruction Register

Bit position:	3	2	1	0
Bit field:	TS[3:0]			
Value after reset:	1	1	1	0

Bit	Symbol	Function	R/W
3:0	TS[3:0]	Test Bit Set The command configuration for these bits TS[3:0] / Instruction: 0x0: EXTEST 0x1: SAMPLE/PRELOAD 0x3: IDCODE (Renesas code) 0x5: CLAMP 0x6: HIGHZ 0xF: BYPASS Others: Reserved	—

JTAG instructions can be transferred to the JTIR register by serial input from the TDI pin. The JTIR register is initialized when a power-on reset occurs, or when the TAP controller is in the Test-Logic-Reset state.

49.2.2 JTIDR : ID Code Register

Bit position:	31																															0
Bit field:	DID[31:0]																															
Value after reset:	0	0	0	0	1	0	0	0	0	1	0	1	1	1	0	1	1	0	1	0	0	1	0	0	0	1	0	0	0	1	1	1

Bit	Symbol	Function	R/W
31:0	DID[31:0]	Device ID These bits store the fixed value that indicates the device IDCODE (0x085D_A447).	—

The JTIDR register data is output from the TDO pin when the IDCODE instruction is executed. After a reset release, the DID[31:0] of JTIDR changes into the Arm® debug code. See the *Arm® CoreSight™ SoC-400 Technical Reference Manual* (ARM DDI 0480F).

49.2.3 JTBPR : Bypass Register

The JTBPR register is a 1-bit register and is connected between the TDI and TDO pins when the JTIR register is set to BYPASS mode. The JTBPR register cannot be read from or written to by the CPU.

49.2.4 JTBSR : Boundary Scan Register

The JTBSR register is a shift register for controlling the external input and output pins of this device, and is distributed across the pads. To apply the JTBSR register in boundary-scan testing, issue the EXTEST, SAMPLE/PRELOAD, CLAMP and HIGHZ instructions. The BSDL file describes the associations between the JTBSR register bits and the pins of this device. The value after reset is undefined.

49.3 Operation

During a reset, the JTAG ports, TCK, TMS, TDI and TDO are assigned as default pin functions. The TCK, TMS and TDI pins are pulled up by the pull-up resistors. Boundary scan testing can be executed after the setup time elapses when POR is negated and RES is driven low.

49.3.1 TAP Controller

Figure 49.2 shows the state transition diagram of the TAP controller. All transitions are controlled by the TMS signal.

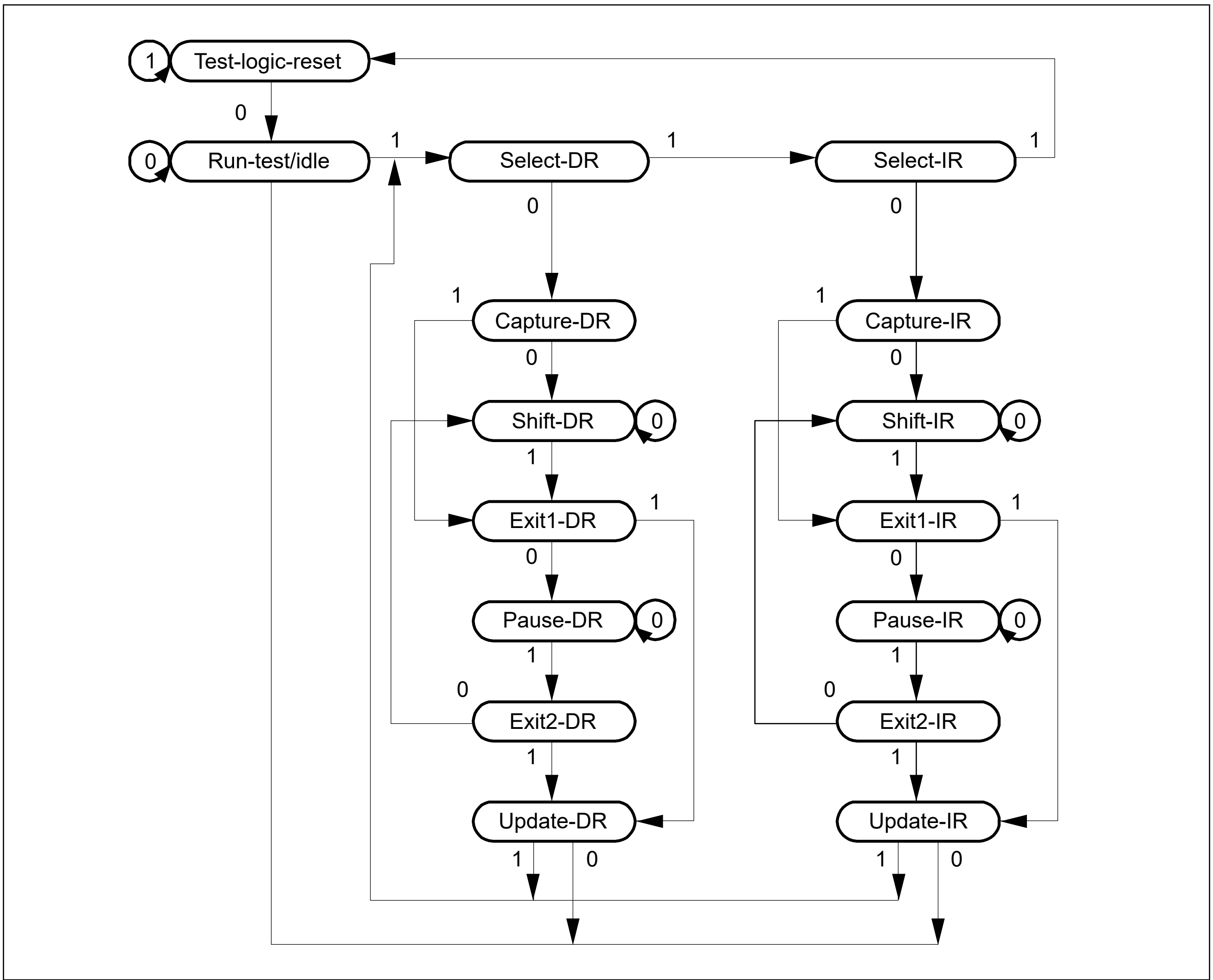

Figure 49.2 State transition diagram of TAP controller

49.3.2 Commands

(1) BYPASS

The BYPASS instruction drives the Bypass Register (JTBPR). This instruction shortens the shift path, facilitating the transfer of serial data to other LSIs on a printed circuit board at higher speeds. While this instruction is being executed, the test circuit has no effect on the system circuits.

The JTBPR register is connected between the TDI and TDO pins. Bypass operation is initiated from the Shift-DR operation. The TDO is low in the first clock cycle in the Shift-DR state. In the subsequent clock cycles, values input to the TDI pin are output from the TDO pin.

(2) EXTEST

The EXTEST instruction is used to test external circuits when this device is installed on the printed circuit board. If this instruction is executed, output pins are used to output test data (specified in the SAMPLE/PRELOAD instruction) from the Boundary Scan Register (JTBSR) to the other devices, and input pins are used to input the test result.

(3) SAMPLE/PRELOAD

The SAMPLE/PRELOAD instruction is used to input data from the internal circuits of this device to the JTBSR register, output data from the scan path, and reload the data to the scan path. While this instruction is executed, input signals are directly input to this device and output signals are also directly output to the external circuits. This device system circuit is not affected by this instruction.

In SAMPLE operation, the JTBSR register latches a snapshot of the data transferred from the input pins to the internal circuit or data transferred from the internal circuit to the output pins. The latched data is read from the scan path. The JTBSR register latches the data snapshot on the rising edge of the TCK pin in the Capture-DR state. The data snapshot is only transferred from the internal circuit to the output pins during a reset.

In PRELOAD operation, the initial value is written from the scan path to the parallel output latch of the JTBSR register prior to the EXTEST instruction execution. If EXTEST is executed without executing this PRELOAD operation, undefined values are output from the beginning to the end (transfer to the output latch) of the EXTEST sequence. (In the EXTEST instruction, output parallel latches are always output to the output pins.)

(4) IDCODE

When the IDCODE instruction is selected, the ID Code Register (JTIDR) value is output to the TDO pin in the Shift-DR state of the TAP controller. In this case, the JTIDR register value is output LSB-first. During this instruction execution, the test circuit does not affect the system circuit.

(5) CLAMP

When the CLAMP instruction is selected, output pins output the JTBSR register value that was specified in the SAMPLE/PRELOAD instruction in advance. While the CLAMP instruction is selected, the status of the JTBSR register is maintained regardless of the TAP controller state.

The JTBPR register is connected between the TDI and TDO pins, leading to the same operation as when the BYPASS instruction is selected.

(6) HIGHZ

When the HIGHZ instruction is selected, all output pins enter high-impedance state. While the HIGHZ instruction is selected, the JTBSR register is maintained regardless of the state of the TAP controller.

The JTBPR register is connected between the TDI and TDO pins, leading to the same operation as when the BYPASS instruction is selected.

49.4 Usage Notes

The boundary scan function is subject to the following constraints:

- The boundary scan must be executed when the RES pin is driven low
- Serial data input/output is in LSB order, as shown in Figure 49.3

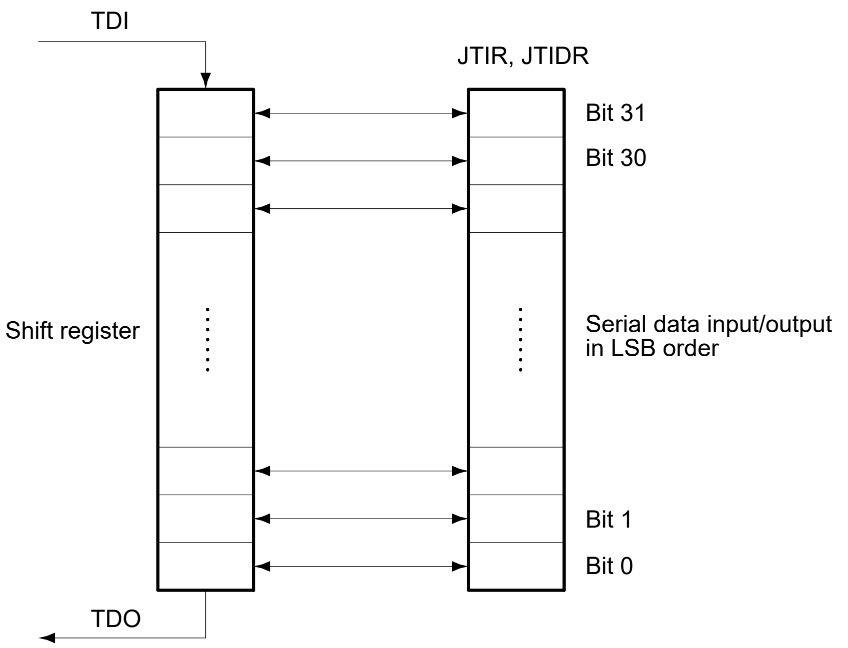

Figure 49.3 Serial data input/output

The following pins cannot be boundary-scanned for Standard Product and SiP Product:

- Power supply pins (VCC, VCC2, VCC_DCDC, VCL, VSS, VSS_DCDC, VBATT, AVCC0, AVSS0, VCC_USB, VSS_USB)
- Analog reference pins (VREFH0, VREFL0, VREFH, VREFL)
- Clock pins (EXTAL, XTAL, XCIN, and XCOUT)
- Reset pin (RES)
- USBFS pin (USB_DP, USB_DM)
- The boundary-scan pins (TCK, TMS, TDI, and TDO).
- The switching regulator pin (VLO)

The following pins cannot be boundary-scanned for SiP Product:

- SiP flash pins (OM_1_SCLK, OM_1_CS1, OM_1_DQS, OM_1_SIO7-0, OM1_RESET and OM_1_ECSINT1)